ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **WINTER SEMESTER, 2024–2025**

DURATION: 2 HOURS **FULL MARKS: 120**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. A system of linear equations is given in matrix form as $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} 1 & 2 & 3 & -2 & -1 \\ 2 & 2 & 12 & -2 & -4 \\ 3 & 4 & 15 & -4 & -5 \\ 1 & 4 & -3 & -3 & 3 \\ -2 & -4 & -6 & 0 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$$

a) Between $\mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 6 \\ -1 \\ -4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -1 \\ 3 \\ 2 \\ -8 \\ 10 \end{bmatrix}$, for which vector $\mathbf{b}$ is the system of linear equations solvable? Provide a justification for your answer. 12 (CO2) (PO1)

b) Find the complete solution to the system $A\mathbf{x} = \mathbf{b}$ by addressing the following: $1+2+4+1+1$ (CO2) (PO1)

i. Identify the **pivot variables** and the **free variables** based on the row-reduced form of the matrix A.

ii. Determine the **particular solution** ($\mathbf{x}_p$) for the vector $\mathbf{b}$ from Question 1.a for which a solution exists.

iii. Determine the **special solution(s)** ($\mathbf{x}_s$) that form the basis of the nullspace.

iv. Express the complete solution as $\mathbf{x}_c = \mathbf{x}_p + \sum c_i \mathbf{x}_{s,i}$, where c_i are the free variables (arbitrary constants) and $\mathbf{x}_{s,i}$ are the special solutions.

v. State the nature of the solution (whether it is always solvable, has a unique solution, or has infinitely many solutions).

c) Is the solution obtained in Question 1.b)iii a vector space? If yes, justify your answer. If no, provide a counterexample to explain why. 3 (CO1) (PO1)

d) The vectors $\mathbf{v}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -10 \\ 4 \\ 0 \\ 4 \\ 1 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} -2 \\ -2 \\ 2 \\ 2 \\ 1 \end{bmatrix}$ cannot form a basis for the Left Nullspace of A, $N(A^T)$. $5+5$ (CO2) (PO1)

i. Provide at least two independent arguments to justify why these vectors cannot represent a basis for $N(A^T)$.

ii. It is guaranteed that one or more vectors from the above set do belong to $N(A^T)$. Determine which of the provided vector(s) form a basis for $N(A^T)$, and justify your answer.

e) Based on your findings in Question 1.b and Question 1.d, answer the following questions regarding the Columnspace $C(A)$, Nullspace $N(A)$, and Left Nullspace $N(A^T)$ of matrix A. For all **three subspaces**: 3 + 3 (CO1) (PO1)

i. Write down the **bases** for each of the three subspaces.

ii. State the **dimension** of each subspace and identify the **vector space** (e.g., $\mathbb{R}^n$) in which it resides.

Solution:

Given, $A = \begin{bmatrix} 1 & 2 & 3 & -2 & -1 \\ 2 & 2 & 12 & -2 & -4 \\ 3 & 4 & 15 & -4 & -5 \\ 1 & 4 & -3 & -3 & 3 \\ -2 & -4 & -6 & 0 & -6 \end{bmatrix}$

So, Augmented Matrix,$[A|b] = \begin{bmatrix} 1 & 2 & 3 & -2 & -1 & | & b_1 \\ 2 & 2 & 12 & -2 & -4 & | & b_2 \\ 3 & 4 & 15 & -4 & -5 & | & b_3 \\ 1 & 4 & -3 & -3 & 3 & | & b_4 \\ -2 & -4 & -6 & 0 & -6 & | & b_5 \end{bmatrix}$

Step 1: $R_2 = R_2 - 2R_1, R_3 = R_3 - 3R_1, R_4 = R_4 - R_1, R_5 = R_5 + 2R_1$

$$\begin{bmatrix} 1 & 2 & 3 & -2 & -1 & | & b_1 \\ 0 & -2 & 6 & 2 & -2 & | & b_2 - 2b_1 \\ 0 & -2 & 6 & 2 & -2 & | & b_3 - 3b_1 \\ 0 & 2 & -6 & -1 & 4 & | & b_4 - b_1 \\ 0 & 0 & 0 & -4 & -8 & | & b_5 + 2b_1 \end{bmatrix}$$

Step 2: $R_3 = R_3 - R_2, R_4 = R_4 + R_2$

$$\begin{bmatrix} 1 & 2 & 3 & -2 & -1 & | & b_1 \\ 0 & -2 & 6 & 2 & -2 & | & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & 0 & | & b_3 - 3b_1 - (b_2 - 2b_1) \\ 0 & 0 & 0 & 1 & 2 & | & b_4 - b_1 + (b_2 - 2b_1) \\ 0 & 0 & 0 & -4 & -8 & | & b_5 + 2b_1 \end{bmatrix}$$

Step 3: $R_5 = R_5 + 4R_4$,Swap $R_3 \longleftrightarrow R_4$

$$\begin{bmatrix} 1 & 2 & 3 & -2 & -1 & | & b_1 \\ 0 & -2 & 6 & 2 & -2 & | & b_2 - 2b_1 \\ 0 & 0 & 0 & 1 & 2 & | & b_4 - b_1 + (b_2 - 2b_1) \\ 0 & 0 & 0 & 0 & 0 & | & b_3 - 3b_1 - (b_2 - 2b_1) \\ 0 & 0 & 0 & 0 & 0 & | & b_5 + 2b_1 + 4(b_4 - b_1 + (b_2 - 2b_1)) \end{bmatrix}$$

Step 4: $R_2 = -R_2/2$

$$\begin{bmatrix} 1 & 2 & 3 & -2 & -1 & | & b_1 \\ 0 & 1 & -3 & -1 & 1 & | & b_1 - (b_2/2) \\ 0 & 0 & 0 & 1 & 2 & | & b_4 - 3b_1 + b_2 \\ 0 & 0 & 0 & 0 & 0 & | & b_3 - b_2 - b_1 \\ 0 & 0 & 0 & 0 & 0 & | & -10b_1 + 4b_2 + 4b_4 + b_5 \end{bmatrix}$$

So, we have obtained the 2 solvability conditions,

1. $b_3 - b_2 - b_1$
2. $-10b_1 + 4b_2 + 4b_4 + b_5$

Let's check the two cases,

Conditions	$b_1 = \{4, 2, 6, -1, -4\}$	$b_2 = \{-1, 3, 2, -8, 10\}$
$b_3 - b_2 - b_1$	0 (satisfied)	-40
$-10b_1 + 4b_2 + 4b_4 + b_5$	0 (satisfied)	0 (satisfied)

Table 1: Checking Solvability Condition

So, $\mathbf{b_2} = \begin{bmatrix} -1 \\ 3 \\ 2 \\ -8 \\ 10 \end{bmatrix}$ satisfies both solvability conditions. [Question 1.a]

Replacing the augmented part with b_2, we get,

$$\begin{bmatrix} 1 & 2 & 3 & -2 & -1 & | & -1 \\ 0 & 1 & -3 & -1 & 1 & | & -5/2 \\ 0 & 0 & 0 & 1 & 2 & | & -2 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Step 5: $R_2 = R_2 + R_3, R_1 = R_1 + 2R_3$

$$\begin{bmatrix} 1 & 2 & 3 & 0 & 3 & | & -5 \\ 0 & 1 & -3 & 0 & 3 & | & -9/2 \\ 0 & 0 & 0 & 1 & 2 & | & -2 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Step 6: $R_1 = R_1 - 2R_2$

$$\begin{bmatrix} 1 & 0 & 9 & 0 & -3 & | & 4 \\ 0 & 1 & -3 & 0 & 3 & | & -9/2 \\ 0 & 0 & 0 & 1 & 2 & | & -2 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$RREF(A)$$

In matrix A, the pivot columns are column 1, 2 and 4, free columns are column 3 and 5. [Question 1.b)i]

$$\begin{bmatrix} 1 & 0 & 9 & 0 & -3 & | & 4 \\ 0 & 1 & -3 & 0 & 3 & | & -\frac{9}{2} \\ 0 & 0 & 0 & 1 & 2 & | & -2 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

So, there will be 2 special solutions (equal to number of free columns),

$$X_{special}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \left\{ \begin{bmatrix} -9 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

(Setting values of free variables as per convention) Question 1.b)iii]
And, the particular solution is,

$$X_{particular}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 4 \\ -9/2 \\ 0 \\ -2 \\ 0 \end{bmatrix}$$

(Setting values of free variables equal to 0 as per convention) Question 1.b)ii]
So, the complete solution is,

$$X_{complete} = X_{particular} + X_{special} = \begin{bmatrix} 4 \\ -9/2 \\ 0 \\ -2 \\ 0 \end{bmatrix} + \left\{ \begin{bmatrix} -9 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

[Question 1.b)iv]
[To check your answer $AX_{particular} = 0$ and $AX_{special} = b_2$]
The solution is **infinite** (because the nullspace is nonzero) if $b_2 \in C(A)$ (the column space of A), which is the case for b_2. [Question 1.b)v]

To answer whether $X_{complete}$ is a vector space or not, $X_{complete}$ is composed of **a single vector** ($X_{particular}$) and a $2D$ space ($X_{special}$),(with no overlaps)
Lets just take a simple vector,
Consider $k_1 X_{particular}$, where k_1 is any scalar. For any value apart from $k_1 = 1$, $k_1 X_{particular}$ can't be contained by $X_{complete}$. For example, for $k_1 = 4$ we get $4 \begin{bmatrix} 4 \\ -9/2 \\ 0 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 16 \\ -18 \\ 0 \\ -8 \\ 0 \end{bmatrix}$. This vector is not contained by $X_{complete}$. So, $X_{complete}$ is **not a vector space**. [Question 1.c]

If dot product of a vector x with a basis of $C(A)$ is 0, it means $x \in C(A)$. The bases of $C(A)$ are columns 1, 2 and 4 or A, we can take any, for now we will be working with the second one $c = \begin{bmatrix} 2 \\ 2 \\ 4 \\ 4 \\ -4 \end{bmatrix}$.

$$c \cdot v_1 = \begin{bmatrix} 2 \\ 2 \\ 4 \\ 4 \\ -4 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = 0 \qquad (Yes)$$

$$c \cdot v_2 = \begin{bmatrix} 2 \\ 2 \\ 4 \\ 4 \\ -4 \end{bmatrix} \cdot \begin{bmatrix} -10 \\ 4 \\ 0 \\ 4 \\ 1 \end{bmatrix} = 0 \qquad (Yes)$$

$$c \cdot v_3 = \begin{bmatrix} 2 \\ 2 \\ 4 \\ 4 \\ -4 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ -2 \\ 2 \\ 2 \\ 1 \end{bmatrix} = 4 \neq 0 \qquad (\mathbf{NO})$$

So, $v_3 \notin N(A^T)$

But if you take the first one $c = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \\ -2 \end{bmatrix}$ (you won't be able to differentiate, your test will fail).

$$c \cdot v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = 0 \qquad (Yes)$$

$$c \cdot v_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} -10 \\ 4 \\ 0 \\ 4 \\ 1 \end{bmatrix} = 0 \qquad (Yes)$$

$$c \cdot v_3 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ -2 \\ 2 \\ 2 \\ 1 \end{bmatrix} = 0 \qquad (Yes)$$

[Question 1.d] You can test it for all the bases of $C(A)$, it won't change the result.

Vector Subspace	Columnspace,$C(A)$	Nullspace,$N(A)$	Left Nullspace,$N(A^T)$
Bases	$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 4 \\ 4 \\ -4 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ -4 \\ -3 \\ 0 \end{bmatrix}$	$\begin{bmatrix} -9 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ 0 \\ -2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -10 \\ 4 \\ 0 \\ 4 \\ 1 \end{bmatrix}$
Subspace of	$\mathbb{R}^5$	$\mathbb{R}^5$	$\mathbb{R}^5$.

Table 2: Bases of $C(A), N(A)$ and $N(A^T)$

To be noted: You don't need to compute $RREF(A^T)$.
Answers:

a) $\mathbf{b_2} = \begin{bmatrix} -1 \\ 3 \\ 2 \\ -8 \\ 10 \end{bmatrix}$ satisfies both solvability conditions.

b) i) In matrix A, the pivot columns are column $1, 2$ and 4, free columns are column 3 and 5.

$$\left[\begin{array}{ccccc|c} 1 & 0 & 9 & 0 & -3 & 4 \\ 0 & 1 & -3 & 0 & 3 & -\frac{9}{2} \\ 0 & 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

ii) The particular solution is,

$$X_{particular}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 4 \\ -9/2 \\ 0 \\ -2 \\ 0 \end{bmatrix}$$

iii) There are 2 special solutions ,

$$X_{special}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \left\{ \begin{bmatrix} -9 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

iv) The complete solution is,

$$X_{complete} = X_{particular} + X_{special} = \begin{bmatrix} 4 \\ -9/2 \\ 0 \\ -2 \\ 0 \end{bmatrix} + \left\{ \begin{bmatrix} -9 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

v) The solution is **infinite** (because the Nullspace is nonzero) if $b_2 \in C(A)$ (the column space of A), which is the case for b_2.

c) No, $X_{complete}$ is not a vector space.

d) i) Two independent arguments to justify why these vectors cannot represent a basis for $N(A^T)$.

- The rank of A is 3 (number of pivot columns), and total number of columns, $n = 5$. So, dimension of the Left Nullspace should be $n - r = 5 - 3 = 2$, but 3 vectors are given, so all of the three vectors can't form the left nullspace, $N(A^T)$.
- The dot product of all three vectors doesn't yield zero with the bases of $C(A)$. (The third vector v_3 fails.)

ii) $v_1(\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}), v_2(\begin{bmatrix} -10 \\ -4 \\ 0 \\ 4 \\ 1 \end{bmatrix})$ forms the bases of $N(A^T)$, verified by calculating the dot product with bases of $C(A)$.

e) Table provided in Table 2.

Rubrics

- 8 marks for correctly determining $RREF(A)$.
- 2 marks (1 mark for each) for correctly determining the solvability conditions.
- 2 marks for correctly choosing b_2 over b_1.
- 1 mark for correctly identifying pivot and free columns. 2 marks for correctly identifying $x_{paticular}$. 4 marks (2 marks for each) for correctly identifying $x_{special}$.
- 1 mark for correctly writing down the complete solution.
- 1 mark for commenting on nature of solution.
- 1 mark for stating that the complete solutions doesn't form a vectorspace.
- 2 marks for justification/counterexample to defend your opinion correctly.
- 4 marks (2 marks for each) for providing 2 arguments supporting the given statement.
- 6 marks for correctly identifying which 2 vectors belong to the Left Nullspace $N(A^T)$.
- 3 marks (1 mark for each) for correctly identifying the bases of the three subspaces ($C(A), N(A)$, and $N(A^T)$).
- 3 marks (1 mark for each) for correctly identifying the subspaces of the three subspaces ($C(A), N(A)$, and $N(A^T)$).

2. Figure 1 illustrates the inflow and outflow of a network containing three nodes: X, Y, and Z. The weights of the directed edges between each pair of nodes are equal. For example, for the nodes X and Y,

$$2a + 7c = 3b - 11$$

.

a) Determine the values of the unknown flows a, b, and c by completing the following steps: $2+1+7+1$ (CO2) (PO1)

i. Derive the system of three linear equations based on the flow conservation principle at nodes X, Y, and Z as shown in Figure 1.

ii. Express the system of equations (obtained in Question 2.a)i) in the matrix form $A\mathbf{x} = \mathbf{b}$, clearly identifying the coefficient matrix A, the variable vector $\mathbf{x}$, and the constant vector $\mathbf{b}$.

iii. Perform Gaussian Elimination on the augmented matrix $[A|\mathbf{b}]$. You must show the

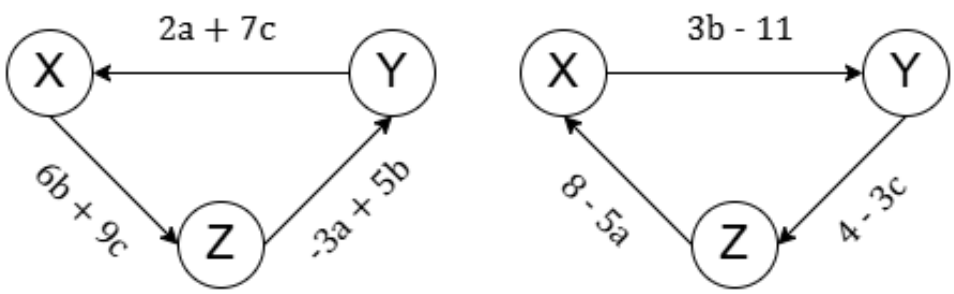

Figure 1: Inflow and Outflow Diagram of a Graph

specific elimination matrices E_{ij} used to zero out the entry at row i and column j. If a pivot is zero, utilize a permutation matrix P_{ij} to perform a row swap.

iv. Determine the final values of a, b, and c using back-substitution on the resulting upper triangular matrix.

b) Factorize the coefficient matrix A from Question 2.a)ii into LDU form, where L is a unit lower triangular matrix (the inverse of the product of elimination matrices, E^{-1}), D is a diagonal matrix containing the pivots, and U is the unit upper triangular matrix obtained by scaling the rows of the matrix from Question 2.a)iii so that all diagonal entries are 1. 6 + 3 (CO1) (PO1)

i. Determine the specific entries for matrices L, D, and U.

ii. Calculate the **trace** and the **determinant** for each of the three matrices: L, D, and U.

Solution:

As per the given scenario, we can construct the following equations based on the diagram and the given scenario:

Nodes	Egde 1	Egde 2	Equation
X & Y	$2a + 7c$	$3b - 11$	$2a - 3b + 7c = -11$
Y & Z	$-3a + 5b$	$4 - 3c$	$-3a + 5b + 3c = 4$
Z & X	$6b + 9c$	$8 - 5a$	$5a + 6b + 9c = 8$

Table 3: Directed Edges between same pair of nodes

Therefore, the obtained equations,

$$\begin{aligned} 2a - 3b + 7c &= -11 \\ -3a + 5b + 3c &= 4 \\ 5a + 6b + 9c &= 8 \end{aligned}$$

[Question 2.a)i]

To represent the given equations in $Ax = b$ form, where A, x and b are coefficient matrix , the variable vector, and the constant vector respectively,

$$\begin{bmatrix} 2 & -3 & 7 \\ -3 & 5 & 3 \\ 5 & 6 & 9 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -11 \\ 4 \\ 8 \end{bmatrix}$$

$$Ax = b$$

[Question 2.a)ii]

$$[A|b] = \left[\begin{array}{ccc|c} 2 & -3 & 7 & -11 \\ -3 & 5 & 3 & 4 \\ 5 & 6 & 9 & 8 \end{array}\right]$$

Step 1: $R_2 = R_2 + 3R_1/2$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\begin{array}{ccc|c} 2 & -3 & 7 & -11 \\ -3 & 5 & 3 & 4 \\ 5 & 6 & 9 & 8 \end{array}\right] = \left[\begin{array}{ccc|c} 2 & -3 & 7 & -11 \\ 0 & 1/2 & 27/2 & -25/2 \\ 5 & 6 & 9 & 8 \end{array}\right]$$

$$E_{21}[A|b] = [A'|b]$$

Step 2: $R_2 = R_2 - 5R_1/2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{5}{2} & 0 & 1 \end{bmatrix} \left[\begin{array}{ccc|c} 2 & -3 & 7 & -11 \\ 0 & 1/2 & 27/2 & -25/2 \\ 5 & 6 & 9 & 8 \end{array}\right] = \left[\begin{array}{ccc|c} 2 & -3 & 7 & -11 \\ 0 & 1/2 & 27/2 & -25/2 \\ 0 & 27/2 & -17/2 & 71/2 \end{array}\right]$$

$$E_{31}[A'|b] = [A''|b]$$

Step 3: $R_3 = R_3 - 27R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -27 & 1 \end{bmatrix} \left[\begin{array}{ccc|c} 2 & -3 & 7 & -11 \\ 0 & 1/2 & 27/2 & -25/2 \\ 0 & 27/2 & -17/2 & 71/2 \end{array}\right] = \left[\begin{array}{ccc|c} 2 & -3 & 7 & -11 \\ 0 & 1/2 & 27/2 & -25/2 \\ 0 & 0 & -373 & 373 \end{array}\right]$$

$$E_{32}[A''|b] = [U|c]$$

[Question 2.a)iii]

So, we obtain,

$$2a - 3b + 7c = -11 \tag{1}$$

$$\frac{b}{2} + \frac{27c}{2} = -\frac{25}{2} \tag{2}$$

$$-373c = 373 \tag{3}$$

From (3),

$$c = -1.$$

Substitute $c = -1$ into (2):

$$\frac{b}{2} + \frac{27(-1)}{2} = -\frac{25}{2}$$

$$\frac{b}{2} - \frac{27}{2} = -\frac{25}{2}$$

$$\frac{b}{2} = 1 \quad \Rightarrow \quad b = 2.$$

Substitute $b = 2$ and $c = -1$ into (1):

$$2a - 3(2) + 7(-1) = -11$$

$$2a - 13 = -11$$

$$2a = 2 \quad \Rightarrow \quad a = 1.$$

$$\boxed{a = 1,\ b = 2,\ c = -1}$$

[Question 2.a)iv]

$$L = E^{-1}$$
$$\Rightarrow\ L = (E_{32}E_{31}E_{21})^{-1}$$
$$\Rightarrow\ L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$$
$$\Rightarrow\ L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{5}{2} & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -27 & 1 \end{bmatrix}^{-1}$$
$$\Rightarrow\ L = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{5}{2} & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 27 & 1 \end{bmatrix}^{-1}$$
$$\Rightarrow\ L = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ \frac{5}{2} & 27 & 1 \end{bmatrix}$$

So,

$$A = LU$$
$$\Rightarrow\ A = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ \frac{5}{2} & 27 & 1 \end{bmatrix} \begin{bmatrix} 2 & -3 & 7 \\ 0 & 1/2 & 27/2 \\ 0 & 0 & -373 \end{bmatrix}$$
$$\Rightarrow\ A = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ \frac{5}{2} & 27 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & -373 \end{bmatrix} \begin{bmatrix} 1 & -\frac{3}{2} & \frac{7}{2} \\ 0 & 1 & 27 \\ 0 & 0 & 1 \end{bmatrix}$$
$$A = LDU$$

[Question 2.b)i]

Properties	L	D	U
Matrix	$\begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ \frac{5}{2} & 27 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & -373 \end{bmatrix}$	$\begin{bmatrix} 1 & -\frac{3}{2} & \frac{7}{2} \\ 0 & 1 & 27 \\ 0 & 0 & 1 \end{bmatrix}$
Trace	$1+1+1=3$	$2+\frac{1}{2}-373=-\frac{741}{2}$	$1+1+1=3$
Determinant	1	$2\cdot\frac{1}{2}\cdot(-373)=-373$	1

Table 4: Properties of the matrices in the LDU decomposition

Rubrics

- 2 marks for correctly forming the equations.
- 1 mark for correctly writing down the equations in matrix form.
- 7 marks (2+2+2+1) for correctly converting $[A|b]$ to $[U|c]$
- 1 marks for correctly determining the values of a, b and c.

- 4 marks for correctly identifying L.
- 2 marks for correctly splitting U into DU.
- 3 marks ($3 \times 0.5 + 3 \times 0.5$) for correctly determining the tract and determinants L, D and U.

3. a) A linear transformation maps the vector $\begin{bmatrix} -8 \\ 2 \end{bmatrix}$ to $\begin{bmatrix} -8 \\ 4 \end{bmatrix}$. The transformation compresses the entire space by a factor of half and also reverses its orientation. The trace of the transformation matrix is 1. 10 (CO2) (PO1)

Find the transformation matrix corresponding to this transformation. In other words, determine the basis vectors (columns) of the transformation matrix.

Solution:

Let the Transformation matrix be T.

Given, T transforms vector $\begin{bmatrix} -8 \\ 2 \end{bmatrix}$ to $\begin{bmatrix} -8 \\ 4 \end{bmatrix}$. So,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} -8 \\ 2 \end{bmatrix} = \begin{bmatrix} -8 \\ 4 \end{bmatrix}$$

And, $det(T) = ad - bc = -\frac{1}{2}$ ($-ve$ sign for reverse, and $\frac{1}{2}$ is the factor by which the space is transformed by T).

Also given, $Trace(A) = a + d = 1$.

Putting all the available information together,

$$-8a + 2b = -8 \tag{4}$$
$$-8c + 2d = 4 \tag{5}$$
$$ad - bc = -\frac{1}{2} \tag{6}$$
$$a + d = 1. \tag{7}$$

From the first equation,

$$-4a + b = -4 \quad \Rightarrow \quad b = 4a - 4.$$

From the second equation,

$$-4c + d = 2 \quad \Rightarrow \quad d = 4c + 2.$$

Substitute $d = 4c + 2$ into $a + d = 1$:

$$a + 4c + 2 = 1 \quad \Rightarrow \quad a = -4c - 1.$$

Now substitute $a = -4c - 1$ into $b = 4a - 4$:

$$b = 4(-4c - 1) - 4 = -16c - 8.$$

Substitute a, b, d into $ad - bc = -\frac{1}{2}$:

$$(-4c - 1)(4c + 2) - (-16c - 8)c = -\frac{1}{2}.$$

Simplifying,

$$-16c^2 - 12c - 2 + 16c^2 + 8c = -\frac{1}{2}$$

$$-4c - 2 = -\frac{1}{2}$$

$$-4c = \frac{3}{2} \quad \Rightarrow \quad c = -\frac{3}{8}.$$

Then

$$a = -4\left(-\tfrac{3}{8}\right) - 1 = \tfrac{1}{2}, \qquad d = 4\left(-\tfrac{3}{8}\right) + 2 = \tfrac{1}{2},$$

$$b = -16\left(-\tfrac{3}{8}\right) - 8 = -2.$$

$$\boxed{a = \tfrac{1}{2}, \quad b = -2, \quad c = -\tfrac{3}{8}, \quad d = \tfrac{1}{2}}$$

So, $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -2 \\ -\frac{3}{8} & \frac{1}{2} \end{bmatrix}$

Rubrics

- 4 marks (1 mark for each) for correctly identifying four the equations.
- 4 marks for correctly solving the equations to find out the transformation matrix.
- 2 marks (4×0.5) for correctly getting each value of transformation matrix right.

b) Let $\vec{p}$, $\vec{q}$, and $\vec{r}$ be three linearly independent vectors. The following regions are defined: 4 + 10 (CO2) (PO1)

$$X = \{\, k_1\vec{p} + k_2\vec{q} + \vec{r} \mid k_1, k_2 \in \mathbb{R} \,\} \quad \text{and} \quad Y = \{\, k_3\vec{r} \mid k_3 \in \mathbb{R} \,\},$$

where k_1, k_2, and k_3 are scalars.

i. Roughly sketch X and Y.

ii. Determine whether the set $X \cup Y$ and the set $X \cap Y$ form vector spaces. Provide proper justification for your answers.

Solution:

The sketches of X and Y are as follows [Question 3.b)ii]:

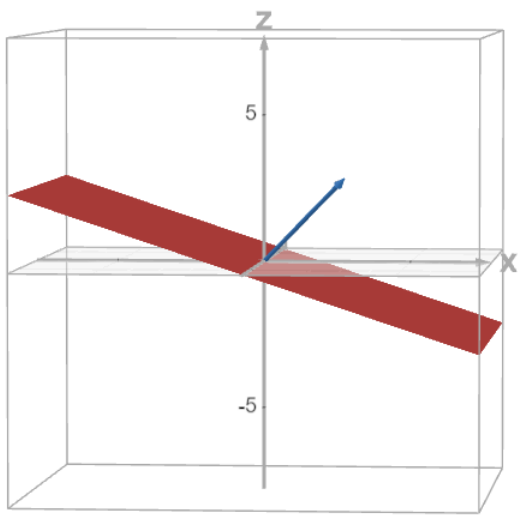

Figure 2: Sketch of X

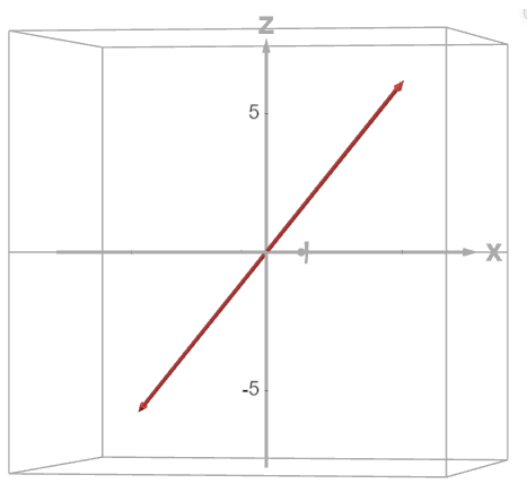

Figure 3: Sketch of Y

$X \cup Y$:

The sets X and Y are given by:

$$X = \{k_1\vec{p} + k_2\vec{q} + \vec{r} \mid k_1, k_2 \in \mathbb{R}\}, \quad Y = \{k_3\vec{r} \mid k_3 \in \mathbb{R}\}.$$

Then

$$X \cup Y = \{k_1\vec{p} + k_2\vec{q} + \vec{r} \mid k_1, k_2 \in \mathbb{R}\} \cup \{k_3\vec{r} \mid k_3 \in \mathbb{R}\}.$$

We show that $X \cup Y$ is *not* a vector space. Take

$$\vec{x} = \vec{r} \in Y, \qquad \vec{y} = \vec{p} + \vec{r} \in X.$$

Then

$$\vec{x} + \vec{y} = \vec{p} + 2\vec{r}.$$

Now, $\vec{p} + 2\vec{r} \notin Y$ since it is not a scalar multiple of $\vec{r}$, and $\vec{p} + 2\vec{r} \notin X$ since every vector in X has the coefficient of $\vec{r}$ equal to 1. Hence

$$\vec{p} + 2\vec{r} \notin X \cup Y,$$

so $X \cup Y$ is not closed under addition. Therefore,

$$\boxed{X \cup Y \text{ is not a vector space.}}$$

$X \cap Y$**:**

The intersection is

$$X \cap Y = \{\vec{r}\}.$$

This set is not a vector space because it does not contain the zero vector, and scalar multiples of $\vec{r}$ (other than $\vec{r}$ itself) are not in the set. Hence,

$$\boxed{X \cap Y \text{ is not a vector space.}}$$

Rubrics:

- 4 marks (2 marks each) for correct rough sketching of X and Y.
- 10 marks (5 marks each) for correctly justifying that $X \cup Y$ and $X \cap Y$ do not form vector spaces.

c) Show that for any permutation matrix P, the transpose of P is equal to its inverse: 6 (CO1) (PO1)

$$P^T = P^{-1}$$

Your argument should be based on the defining properties of permutation matrices and the effect of transposition and inversion on them.

Solution:

Claim: For any permutation matrix P,

$$P^T = P^{-1}.$$

Proof:

A permutation matrix P is obtained by permuting the rows of the identity matrix. By definition, P has exactly one entry equal to 1 in each row and each column, and all other entries are 0.

Consider the product $P^T P$. The (i, j)-entry of this product is given by the dot product of the i-th row of P^T (which is the i-th column of P) with the j-th column of P. Since each column of P contains exactly one 1, these dot products satisfy

$$(P^T P)_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

Hence,

$$P^T P = I.$$

Similarly, one can show that $PP^T = I$. Therefore, P^T is the inverse of P, and we conclude

$$\boxed{P^T = P^{-1}}.$$

Example:
Let

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then

$$P^T = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Computing the product,

$$P^T P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Thus, in this example, $P^T = P^{-1}$, which illustrates the general result. (It can be also shown with respect to a matrix A that $AP^T = AT^{-1}$)
Rubrics:

- 3 marks for correct logical reasoning.
- 3 marks for providing example or proving the claim mathematically.

4. a) There are n students at a renowned International University. Their main occupation is forming various societies every month so that students can become Presidents, Secretaries, or occupy made up positions on impressive sounding Executive Committees. After a while, this started to get boring, and the university began losing its aura. In order to limit the number of societies, the *ExcludedFromEC* Council decreed the following innocent-looking rules: 5 + 10 (CO2) (PO1)

- Each society has to have an **odd number** of members.
- Every **pair of societies** must have an **even** number of members in **common**.

Show that under these rules, it is impossible to form more societies than n, the number of students. Attempt to solve this problem by approaching it in the following way:

i. Let the number of students be n (labeled $1, 2, \dots, n$), and let there be m societies $S_1, S_2, \dots, S_m$. Construct an $m \times n$ incidence matrix $A = (a_{ij})$, where

$$a_{ij} = \begin{cases} 1, & \text{if } j \in S_i \text{ (Student } j \text{ is in Society } i\text{)}, \\ 0, & \text{otherwise.} \end{cases}$$

(In other words, construct a matrix with societies along rows and students along columns, if a student belongs to a society put 1, otherwise 0). Construct this matrix for small values of n and m of your choice, making sure that $n > 3$, $m > 3$, and $n \neq m$.

ii. Discuss your solution approach. What would be sufficient to show that it is not possible to form more than n societies under the new rules? If possible, provide the proof.

Solution:

To satisfy the rules given by the *ExcludedFromEC* with $n = 5$ students and $m = 4$ societies, we consider a university with $n = 5$ **students** and $m = 4$ **societies**. We define the societies such that each has an odd number of members and every pair of societies has an even number of members in common. Furthermore, we ensure that every student belongs to at least one society.
Let the students be labeled $\{1, 2, 3, 4, 5\}$. We define the societies as follows:

- **Society 1:** Student 1

- **Society 2:** Student 2
- **Society 3:** Student 3
- **Society 4:** Student 4

Here, each society has exactly one member (odd), and any pair of societies has zero students in common (even). Student 5 is not part of any society in this example, but it does not violate any rules.

Incidence Matrix

The incidence matrix A, where rows represent societies and columns represent students, is:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

[Question 4.a)i (can also be shown with any other valid matrix)]

$$AA^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I_3$$

- Diagonal entries $= 1$ correspond to odd-sized societies.
- Off-diagonal entries $= 0$ correspond to even intersections between societies.

To argue why it is **not possible to form more than n societies** under these rules, consider the following:

- Each society corresponds to a row vector in $\mathbb{F}_2^n$, with an odd number of ones.
- Every pair of societies has an even number of members in common, so the dot product of any two distinct rows is 0 in $\mathbb{F}_2$.
- Therefore, the rows of A are **mutually orthogonal** over $\mathbb{F}_2$.
- It is not necessary to provide a complete proof. Noting that the rank of A (or equivalently AA^T) cannot exceed n is enough to receive a substantial portion of the marks.

Mutual orthogonality in $\mathbb{F}_2^n$ implies that the set of row vectors is linearly independent. Since the dimension of $\mathbb{F}_2^n$ is n, we cannot have more than n linearly independent vectors. Hence, the number of societies m satisfies:

$$m \leq \operatorname{rank}(A) \leq n.$$

Rubrics:

- 5 marks for correctly constructing the matrix.
- 10 marks for logical reasoning.

b) Given, 7 (CO1) (PO1)

$$S = \left\{ \begin{bmatrix} x^3 \\ y^9 \end{bmatrix} \middle| \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \right\}$$

Is S a vector space? If yes, justify your answer. If no, provide a counterexample to demonstrate why it is not.

Solution:

We are given

$$S = \left\{ \begin{bmatrix} x^3 \\ y^9 \end{bmatrix} \middle| \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \right\}.$$

Step 1: Zero vector

Choosing $x = 0$ and $y = 0$ gives

$$\begin{bmatrix} 0^3 \\ 0^9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in S,$$

so the zero vector is in S.

Step 2: Standard basis vectors and negatives

- Choosing $x = 1, y = 0$ gives $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \in S$. Choosing $x = -1, y = 0$ gives $\begin{bmatrix} -1 \\ 0 \end{bmatrix} \in S$.
- Choosing $x = 0, y = 1$ gives $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \in S$. Choosing $x = 0, y = -1$ gives $\begin{bmatrix} 0 \\ -1 \end{bmatrix} \in S$.

Thus, S contains the standard basis vectors $[1, 0]$, $[0, 1]$ and their negatives.

Step 3: Reaching any vector in $\mathbb{R}^2$

Take any vector $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$.

- To get the first component a, choose $x = \sqrt[3]{a}$ so that $x^3 = a$.
- To get the second component b, choose $y = \sqrt[9]{b}$ so that $y^9 = b$.

Then

$$\begin{bmatrix} x^3 \\ y^9 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} \in S.$$

Step 4: Closure under scalar multiplication and addition

- Negative numbers are allowed since odd roots exist for all real numbers.
- Any linear combination of vectors in S can be represented by choosing appropriate x and y, as shown above.

Since S contains the zero vector, the standard basis vectors, their negatives, and allows any linear combination via appropriate choice of x and y, we conclude:

$$\boxed{S \text{ is a vector space and spans } \mathbb{R}^2.}$$

Rubrics:

- 2 marks for mentioning that it contains zero vector.
- 5 marks for establishing that it is a vectorspace.

c) There are m rows and n columns in a matrix M. The dimension (number of basis vectors) of its nullspace is $n - r$. 3 + 5 (CO1) (PO1)

i. What is the span of the columns of the stated matrix M? (e.g., nothing, line, plane, or specific vector space $\mathbb{R}^k$).

ii. If the matrix is square and its rank is the same as the number of columns, describe the nullspace of the matrix and comment on the span of its columns.

Solution:

The dimension of its Nullspace,$N(A) = n - r$.
So, the dimension of its Columnspace, $C(A) = n - (n - r) = r$. (Since, $Dim(C(A)) + Dim(N(A)) = n$).
So, the span of its Columnspace $C(A)$ will be $\mathbb{R}^r$. [Question 4.c)i]

Given, the matrix is square and its rank is same as the number of columns(n).
So, $n = m = r$. If $Dim(C(A)) = r = n$.
$Dim(N(A)) = n - Dim(C(A)) = n - r = n - n = 0$.
Thus, its Nullspace will only be the **zero vector**.[Question 4.c)ii]

Rubrics:

- 3 marks for correctly identifying the span of its columns as $\mathbb{R}^r$.
- 5 marks for determining that its Nullspace would only contain the zero vector.

MARKS SUMMARY TABLE

Question	Marks	Keywords
1.a	12	Solvability Condition
1.b	9	Complete Solution
1.c	3	Vector Space
1.d	10	Fundamental Subspaces
1.e	6	Linear Equations
2.')i	11	System Modeling
2.a	9	LDU Factorization
3.a	10	Linear Transformation
3.a)i	14	Abstract Vector Spaces
3.b	6	Permutation Matrices
4.a	15	Oddtown Theorem
4.b	7	Not a Vectorspace
4.c	8	Rank and Nullity
Total	120	